Kavish Patel

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SKILLS

Infrastructure & Orchestration: Docker, Kubernetes, Apache Kafka, Apache Spark, Airflow, AWS
AI & ML Systems: PyTorch, TensorFlow, LangChain, LLMs, RAG, Agentic AI Workflows, Vector Search (pgvector)
Data & Distributed Processing: Streaming Pipelines, A/B Testing, Causal Inference, Recommender Systems
Languages & APIs: Python, SQL, R, PostgreSQL, FastAPI, REST

PROJECTS

Containerized Real Time Graph Analytics Pipeline

- Orchestrated a production style Docker and Kubernetes cluster on Minikube integrating Kafka, Zookeeper, and Neo4j, standing up a multi service ingestion and processing system with independently scaled components.
- Implemented PageRank and BFS graph algorithms over streaming NYC Yellow Cab data, tuning throughput and resource allocation across the cluster for near real time ranking and traversal.

Autonomous EDA Agent

- Built a multi agent system with LangGraph (Planner, RTL Generator, Compiler, Simulation, Debugger, Optimizer, Reporter) that converts natural language hardware specs into synthesizable Verilog and iterates autonomously until validation passes.
- Integrated open source EDA tooling, Verilator, Icarus Verilog, Yosys, and OpenROAD, for linting, simulation, gate level synthesis, and static timing analysis.
- Implemented a primary and backup local model escalation strategy (Qwen2.5 Coder 14B escalating to Gemma 3 12B or DeepSeek Coder) so the agent automatically repairs compilation and logic errors.
- Shipped a Dockerized FastAPI backend and Streamlit dashboard for workflow visualization and report generation.

CLIP Guided Diffusion Model on Tiny ImageNet

- Trained a latent diffusion model steered by CLIP ViT-B/32 to generate semantically aligned 64×64 images, running the full training lifecycle: data preparation, GPU training, evaluation, and optimization.
- Reduced inference time by 30% through UNet architecture tuning and optimized sampling steps, balancing compute cost against image fidelity.

INTERNSHIPS

Exxat Inc. Jan 2024 – June 2024
Product Engineer – Client Implementation *Remote, NJ, USA*

- Containerized and deployed an NLP classification model with Docker for production use, sizing the deployment to handle real time query traffic over 10 million records.
- Built automated ML inference pipelines that cut report generation time by 15%, adding monitoring and feedback loops to keep pipeline output and reliability measurable over time.
- Built a Python and Selenium automation system that lifted operational efficiency by 20%, integrating it into cross functional data infrastructure.
- Designed 12 HIPAA and FERPA compliant Power BI dashboards giving academic administrators visibility into student outcomes and compliance metrics.

Accelleron April 2023 – Oct 2023
Data Analyst – Operations *Remote, Switzerland*

- Applied statistical modeling and causal analysis across three years of financial data, surfacing performance trends that informed strategic decisions for 33 regional markets.
- Automated creation and distribution of performance reports for 33 regions in Python, improving data accessibility and enabling real time updates for stakeholders across sales teams.
- Partnered with regional sales managers to define conversion metrics, building dashboards that surfaced actionable signals from the underlying data pipeline.
- Optimized reporting pipelines to reduce manual intervention and preserve data accuracy through a full migration from Tableau to Power BI.

EDUCATION

Master of Science in Data Science, Analytics and Engineering Aug 2024 – May 2026
Arizona State University GPA: 3.7

Bachelor of Engineering in Electronics and Communication - Minor: Data Science 2020 – 2024
Charutar Vidya Mandal University, India GPA: 9.16/10